



Vantage® Research Ultrasound System

Vantage System Initial Setup Instructions

D00036 Rev. B

Summary

This document provides important safety warnings and instructions for the initial setup of the Vantage System.

Table Of Contents

SUMMARY..... 2

1. IMPORTANT NOTES AND WARNINGS 3

2. UNPACK THE VANTAGE SYSTEM 4

3. INITIAL SETUP OF THE VANTAGE HARDWARE SYSTEM..... 5

4. FIRST-TIME STARTUP PROCEDURE..... 7

 4.1 POWER ON THE VANTAGE SYSTEM..... 7

 4.2 LOGIN TO THE SYSTEM..... 7

 4.3 INSTALL THE MATLAB® SOFTWARE..... 7

 4.4 ACTIVATE THE VANTAGE SOFTWARE..... 7

1. Important Notes and Warnings

This section covers important safety warnings that the user should be aware of before using the Vantage System.



Voltage Warning: The Vantage System contains hazardous voltages and can produce other emissions such as heat or acoustic output that can cause injury to the operator or other bystanders. Please use caution.

Note that there are no user-serviceable components inside the chassis. *Do not remove the side covers.*



Environmental Warning: Ensure that the operating environment in which the Vantage System is used conforms to the system specifications in terms of ambient temperature, AC line power (voltage, frequency, and stability), cooling airflow restrictions, mechanical shock, and vibration. Using the system outside these operating limits may result in hazardous conditions.



Noise Warning: The cooling fans in the Vantage hardware system, host controller, and other devices included in an investigational system may lead to a combined audible noise level that could be annoying or distracting to the operator.



Weight Warning - Use caution when lifting the system; it weighs up to 44 lb. (20 Kg.), and a system with a HIFU option can weigh up to 57 lb. (26 Kg.) Before lifting the system, ensure that all cables have been disconnected.



Operator Training Required: The system user interface is designed for flexibility and ease of use in the research environment and differs significantly from the user interface provided by typical commercial ultrasound systems designed for clinical environments.



Do not discard the shipping carton. When transporting the system, always use the shipping carton and foam inserts provided by Verasonics. The box is designed to protect the system when transporting and has handles for easy lifting.

2. Unpack the Vantage System

Use the following guidelines to unpack the Vantage System:

1. Locate an appropriate place for setting up the Vantage System in your lab. Ensure that the location is clean and well-ventilated.
2. After selecting a suitable location, carry the shipping carton and place it next to the table, on which the Vantage System and host controller can be set up for use.
Note: It is recommended that two people lift the system to carry it to the selected location. Do not slide the system or lift the system using the UTA connectors to ensure the integrity of the internal components.
3. Before unpacking the system, examine the shipping carton for any signs of damage—such as tears, gouges, or watermarks. Note any damage to the packaging. If the damage is severe, contact the [Verasonics support team](#). All shipping materials must be preserved in case the system needs to be sent back to Verasonics.
Note: There is an additional cost for replacing the shipping carton and foam inserts.
4. Using a box cutter, open the packaging by releasing any plastic restraints and tape. Do not cut the shipping box or internal packaging.
5. Ensure that the packaging materials are not damaged while opening the box. The Vantage System is held in place by foam inserts on both sides. Examine the foam inserts for any sign of damage.
6. Use a two-person lifting process to place the Vantage System on the table. Lift the Vantage System (along with the foam inserts) vertically out of the box and place the system on the table.
7. After placing the system on the table, carefully remove the foam insert from one end of the Vantage System while supporting the system. Remove the foam insert from the other end of the Vantage System.
8. After examining the foam inserts for any signs of damage, place them in the shipping carton.
9. Save the shipping carton so that it can be reused to pack the Vantage System if you need to ship the system back to Verasonics for repairs.
10. After unpacking the Vantage System, examine the Vantage System for any signs of damage. If any damage is noted, please document the damage using pictures or notes and report it to support@verasonics.com.

If you ordered a host controller from Verasonics, it is packaged in a smaller box included with the system. Repeat the above steps for unpacking the host controller.

While installing the system, ensure that there is a minimum 6-inch clearance on all sides, including the front and rear, to ensure adequate airflow and to prevent excessive over-bending of the optical PCIe cable.

3. Initial Setup of the Vantage Hardware System

After unpacking the Vantage Hardware System, use the following procedure for the initial setup of the system:

1. Place the Vantage Hardware System and the host controller such that there is at least six inches of spacing between them to allow for adequate airflow. In addition, provide adequate space behind the rear end of the system to avoid over-bending of the PCIe cable.

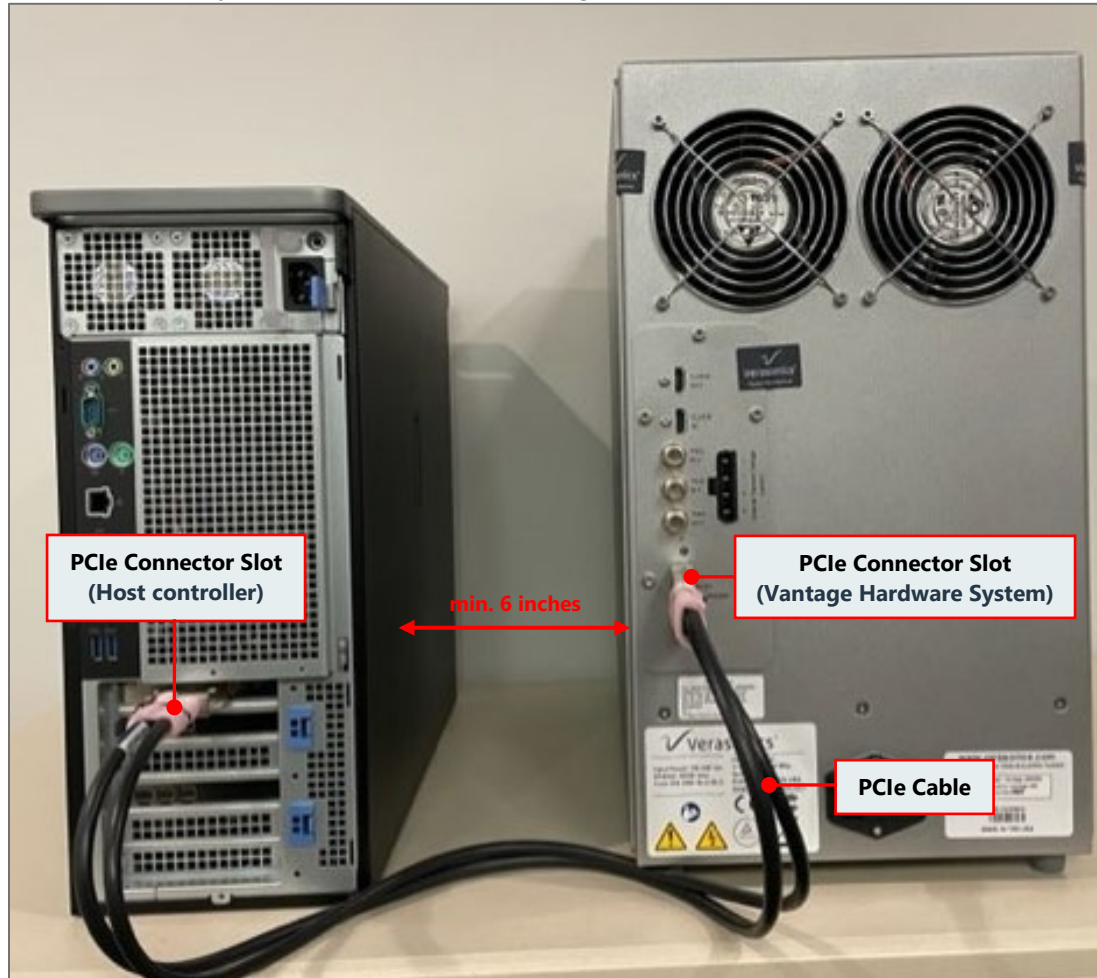


Figure 1: PCIe connection between host controller and Vantage Hardware System

2. Plug one end of the PCIe cable connector into the receptacle located at the back of the Vantage Hardware System.
3. Plug the other end of the PCIe cable into the receptacle located at the back of the host controller.
4. Ensure that the PCIe cable connector is seated properly in the receptacle on both the host controller and the Vantage Hardware System. You should hear a distinct clicking sound when the PCIe cable connector is secured into the receptacle. There should be no gap between the PCIe cable connector and the receptacle.

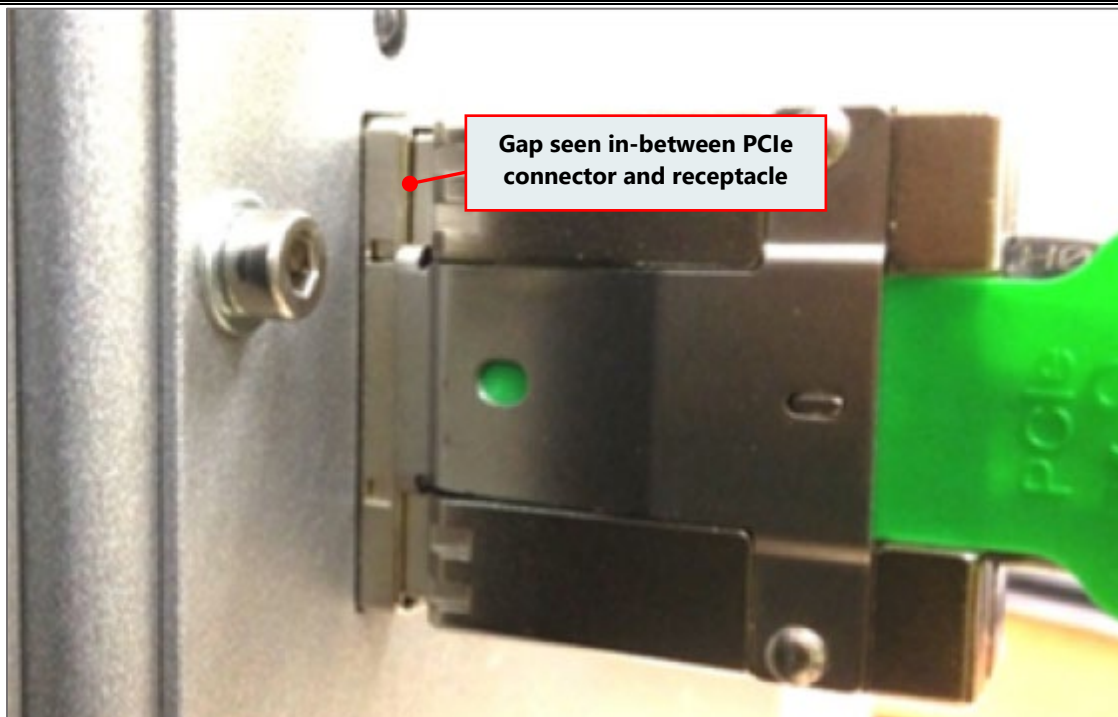


Figure 2: Incorrect PCIe Connection - Gap between PCIe connector and receptacle

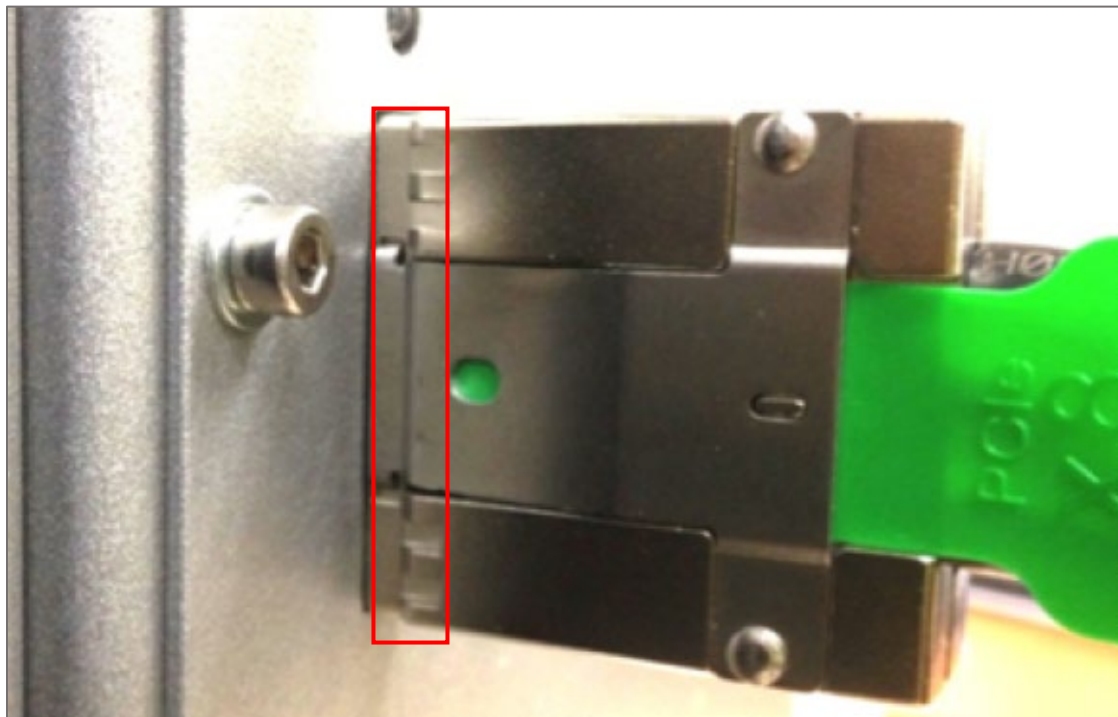


Figure 3: Correct PCIe Connection - No gap between PCIe connector and receptacle

5. Connect the mouse, keyboard, monitor, and other peripheral devices to the host controller.
6. After connecting the peripheral devices, plug in the host controller, Vantage Hardware System, and monitor to a power outlet.

Note: All three units should be connected to the same circuit. Ensure that the circuit is rated to provide adequate power supply to all connected devices. Avoid connecting other devices—it is essential not to overload the circuit.

4. First-time Startup Procedure

After the initial setup of the Vantage System, follow the instructions in this section to start up the system.

4.1 Power On the Vantage System

Use the following steps to power on the Vantage System:

1. Power off the Vantage Hardware System and the host controller.
2. Power on the host controller. The Vantage Hardware System detects the power signal through the PCIe cable and then fully powers up the hardware.
3. If the Vantage Hardware System does not turn on:
 - a. Power off the host controller.
 - b. Check the connection between the Vantage Hardware System and the host controller.
 - c. Ensure that the power switch on the Vantage Hardware System is in the ON position and power on the host controller.
 - d. If the Vantage Hardware System does not power on, contact the Verasonics Support team (email support@verasonics.com).

4.2 Login to the System

After the host controller is powered on and starts up, it displays the Windows login screen.

If you purchased a host controller from Verasonics, enter the following login information:

Username: Verasonics

Password: verasonics

If you are using a third-party host controller, enter the appropriate login information.

Note: Do not add additional users to the host controller—this can create issues with the Vantage Software. If you purchased a Verasonics-supplied host controller, all users should use the Verasonics user account that is configured on the system. For a customer-supplied host controller, it is highly recommended to use a single-user account for all users.

After logging in to the host controller, download a PDF viewer such as Adobe Acrobat or a similar program—this is needed to access the Vantage System documentation.

4.3 Install the MATLAB® Software

On a customer-supplied host controller, download and install MATLAB from the [MathWorks® website](https://www.mathworks.com). To determine the MATLAB version that is compatible with the current Vantage software release, refer to the *Vantage System User Manual* (located in the *Documentation* folder within the *Vantage project* folder).

Note: On a Verasonics-supplied host controller, MATLAB software is preinstalled. The user can skip the installation step; however, since Verasonics does not provide MATLAB licenses, the user must obtain their license.

Activate your MATLAB with Signal Processing Toolbox™ license with MathWorks (<https://www.mathworks.com>). For any issues associated with activating the MATLAB license, contact the MathWorks support team.

4.4 Activate the Vantage Software

On a Verasonics-supplied host controller, Vantage software is preinstalled. The Vantage Project folder is saved at the following file path: PC>Documents>Vantage-X.X.X.

On a customer-supplied host controller, the customer should download the Vantage software distribution and the license file. The latest Vantage software distribution is available in the *Vantage Releases* folder within the *Verasonics Customer Download* folder.

To activate the Vantage Software:

1. Once the MATLAB license has been successfully activated, start MATLAB, and select the *Vantage project* folder as the root directory. If you are using a third-party host controller, navigate to the directory where Vantage Software is installed and select the *Vantage project* folder as the root directory.
2. In the Command Window, type **activate**, and press ENTER. Alternatively, right-click on the 'activate.p' entry on the left-hand side of the MATLAB screen and select "Run".
3. When you run the **activate** command, the software prompts you to accept the EULA agreement. You must accept the agreement to continue.
4. After activating the software, type **VVT** in the Command Window and press ENTER.
5. On the GUI window that pops up, click the **Vantage System Test** button. Do not select other options in the popup window.
 - If any test fails, the "**TEST FAILED**" message appears. Contact the Verasonics support team at support@verasonics.com.
6. Follow and respond to the prompt(s) that appear in the MATLAB Command Window. The test takes a few minutes to run; if the final result "**ALL TESTS PASS**" is displayed in the Command Window, your setup is complete.

After completing the tests, navigate to the subfolder HardwareTest/TestResultLogs to find VVT report files. There are two output files generated prefixed in the format VVT_HOSTNAME_YYMMDD_HHMMSS—one of the files has the suffix "_report.txt" and the other file has the suffix ".log".

Send the two output files as an attachment to support@verasonics.com to let us know that the setup is successful.

Once you've completed the steps above, the Vantage System is ready to use! We recommend signing up for access to the Verasonics Community Portal, verasonicscommunity.com. The community portal provides training videos, app notes, scripts, and other documents—these tools can help you and your team use the Vantage System effectively. In addition, we recommend referring to the *Vantage System User Manual*, *Vantage Sequence Programming Tutorial*, and the *Vantage Release Notes*, all of which can be found in the *Documentation* folder located in the *Vantage Project* folder.

We wish you luck with your experiments!